

Supplementary Revision Addendum

Trustworthy Medication Recommendation: A Safety-Centered Survey of Knowledge-Grounded and LLM-Assisted Methods

Targeted revision update: 3 October 2026

R1. Scope and study-count correction

This addendum supersedes the included-study roster and appraisal totals in the earlier Supplementary File S1. It preserves the original search documentation; it does not assert a new exhaustive database search or reconstruct unavailable screening logs. The original manuscript reports 132 identified records, 39 duplicates, 93 screened records, 37 title/abstract exclusions, and 56 retained full texts for the search through 20 May 2026.

The original roster includes MetaCare++ [43], which predicts diagnoses rather than medications. The earlier S1 itself marks its MR appraisal as not applicable. Removing this entry leaves 55 studies carried forward from the original roster. Five absent full-text studies are added through the targeted revision route, producing a 60-study updated qualitative synthesis. This update includes previously missed pre-cutoff papers as well as later publications; publication coverage and original database-search coverage are distinct.

GAMENet [39], SafeDrug [51], and G-BERT [38] remain pre-2022 background references. They do not enter the 60-study register or the 47-study appraisal subset. The current roster does not count the other eight 2026 references previously added for narrative coverage as screened studies. Family examples and proposed frameworks in the manuscript are illustrative, rather than corpus-frequency statistics.

R2. Targeted full-text additions and contextual records

The following five studies meet the existing patient-specific medication or sequential-regimen selection criterion and have full-text evidence available for this update:

- RpNet [15]: uploaded publisher PDF, 12 pages; Neural Networks 198 (2026), 108618. Online publication: 21 January 2026.
- Nephrotoxic medication management RL [11]: uploaded publisher PDF, 15 pages; Biomedical Signal Processing and Control 112 (2026), 108577. First available online: 20 August 2025. It is within the original date window despite its 2026 volume year.
- MetaDrug [7]: open author manuscript arXiv:2601.22820v1, accessed 3 October 2026; ICDE 2026 status confirmed by the author’s publication list.
- HypeMed [58]: open 28-page author PDF at <https://arxiv.org/pdf/2603.18459>, accessed 3 October 2026; TOIS DOI verified. The inspected PDF includes eICU, unlike an earlier inspected HTML representation; extraction follows the PDF and identifies the absent eICU DDI evaluation.

- GenRxR [10]: open 9-page author PDF at <https://arxiv.org/pdf/2607.24829>, accessed 3 October 2026; RecSys 2026 program and DOI verified. Extraction includes the licensed expert’s synthetic-data plausibility test.

RxGuard [45] is contextual because the primary repository abstract and publication metadata were accessible but the full text was not retrieved for extraction. The blockchain federated-learning paper [1] is contextual because its mixed classification/recommendation tasks do not provide comparable encounter-level medication-set evaluation. Neither record enters the 60-study register. KEHGCN [60] and FLAME [6] are already in the original roster and are not duplicated. IEEE document 11470461 remains unidentified and is not cited on an assumed identity.

R3. Updated included-study register

Table R1 contains 60 studies. O = carried forward from the original S1 after the scope correction; U = targeted full-text addition. Original-roster metadata are carried forward unless explicitly corrected, rather than presented as newly verified full-text extraction. MoleRec’s venue is corrected to WWW 2023; MedRec and HKRec use final issue years 2023 and 2025, respectively, rather than the earlier S1 years. These corrections do not change eligibility. FLAME’s open manuscript also reports MIMIC-IV and eICU generalization tests and uses parsed clinical notes. The machine-readable file `updated_study_register.csv` includes stable study IDs, BibTeX keys, original S1 row numbers, route, metadata source, and appraisal availability. These IDs remain stable when reference numbers change.

Table R1: Updated qualitative-synthesis register (60 studies).

ID	Study/model	Venue (year)	Route	Dataset metadata
MR001	PharmaLLM [2]	HCIS (2024)	O	Private
MR002	REFINE [3]	NeurIPS (2023)	O	MIMIC-III
MR003	Confidence calibration (Chen et al.) [4]	KDD (2025)	O	MIMIC-III + MIMIC-IV
MR004	Context-aware MR (Chen et al.) [5]	AAAI (2023)	O	MIMIC-III
MR005	FLAME [6]	NeurIPS (2025)	O	MIMIC-III; MIMIC-IV and eICU generalization tests
MR006	GPSRec [8]	CIKM (2025)	O	MIMIC-III + MIMIC-IV
MR007	KRAM [9]	ESWA (2026)	O	MIMIC-III
MR008	HI-DR [12]	AAAI (2025)	O	MIMIC-III + MIMIC-IV
MR009	VITA [13]	AAAI (2024)	O	MIMIC-III + MIMIC-IV
MR010	StratMed [14]	arXiv (2023)	O	MIMIC-III
MR011	CausalMed [16]	CIKM (2024)	O	MIMIC-III
MR012	Trans-GAHNet [17]	ESWA (2024)	O	MIMIC-IV
MR013	DGCL [18]	JBH (2023)	O	MIMIC-III
MR014	MR-DTR [19]	WWW (2025)	O	MIMIC-III + MIMIC-IV
MR015	CIDGMed [20]	KBS (2025)	O	MIMIC-III
MR016	TreatRAG [21]	RecSys (2025)	O	MIMIC-IV
MR017	EDRMM [22]	BMC Bioinf. (2025)	O	MIMIC-III + MIMIC-IV
MR018	DNMDR [23]	KBS (2025)	O	MIMIC-III + MIMIC-IV
MR019	IMDR [24]	KDD (2025)	O	MIMIC-III + MIMIC-IV
MR020	TEMPT [25]	TOIS (2025)	O	MIMIC-III + MIMIC-IV + eICU
MR021	LEADER [26]	arXiv (2024)	O	MIMIC-IV
MR022	SHAPE [27]	JBH (2023)	O	MIMIC-III
MR023	DKINet [28]	arXiv (2023)	O	MIMIC-III + MIMIC-IV + eICU
MR024	MedAlign [29]	ACM MM (2025)	O	MIMIC-IV + MIMIC-CXR
MR025	PAUP [30]	IP&M (2026)	O	MIMIC-III
MR026	MedGCN [31]	JBH (2022)	O	MIMIC-III + Private
MR027	ACDNet [32]	JBH (2024)	O	MIMIC-III

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Table R1 continued

ID	Study/model	Venue (year)	Route	Dataset metadata
MR028	BiMoRec [33]	ESWA (2025)	O	MIMIC-III
MR029	KindMed [34]	arXiv (2023)	O	MIMIC-III
MR030	EGNet [35]	Appl. Intell. (2025)	O	MIMIC-III
MR031	KATMed [36]	JBH (2026)	O	MIMIC-III + MIMIC-IV
MR032	RAG-CPMF [37]	Pharm. Res. (2025)	O	Private
MR033	TAHDNet [40]	JBH (2022)	O	MIMIC-III
MR034	CEHMR [41]	AIM (2023)	O	MIMIC-III
MR035	4SDrug [42]	KDD (2022)	O	MIMIC-III
MR036	CAMeR [44]	ICONIP (2023)	O	MIMIC-III
MR037	EXCERF [46]	KBS (2025)	O	MIMIC-III + MIMIC-IV
MR038	ARMR [47]	IJCAI (2025)	O	MIMIC-III + MIMIC-IV
MR039	TAKECare [48]	Inf. Fusion (2025)	O	MIMIC-III
MR040	PROMISE [49]	IP&M (2024)	O	MIMIC-III
MR041	COGNet [50]	WWW (2022)	O	MIMIC-III
MR042	SDRBT [52]	JBH (2025)	O	MIMIC-III
MR043	MoleRec [53]	WWW (2023)	O	MIMIC-III
MR044	AKA-SafeMed [54]	Inf. Sci. (2024)	O	MIMIC-III
MR045	SSPNet [55]	IJCAI (2025)	O	MIMIC-III + MIMIC-IV
MR046	HKRec [56]	TCSS (2025)	O	MIMIC-III
MR047	KEDRec-LM [57]	arXiv (2025)	O	MIMIC-IV
MR048	MedRec [59]	TOIS (2023)	O	MIMIC-III
MR049	KEHGCN [60]	AAAI (2026)	O	MIMIC-III + MIMIC-IV + eICU
MR050	BH3-MedRec [61]	TIST (2026)	O	MIMIC-III + MIMIC-IV + eICU
MR051	LAMO [62]	arXiv (2025)	O	MIMIC-III + MIMIC-IV
MR052	RAREMed [63]	SIGIR (2024)	O	MIMIC-III + MIMIC-IV
MR053	MKAF-Net [64]	Appl. Intell. (2025)	O	MIMIC-III
MR054	TCM-KLLaMA [65]	CBM (2025)	O	Private (TCM)
MR055	DAI-Net [66]	JBH (2024)	O	MIMIC-III + MIMIC-IV + eICU
MR056	RPNet [15]	Neural Networks (2026)	U	MIMIC-III + MIMIC-IV
MR057	Nephrotoxic medication RL [11]	BSPC (online 2025) (2026)	U	MIMIC-IV v3.1; same-hospital later cohort
MR058	MetaDrug [7]	ICDE (2026)	U	MIMIC-III + Kansas AKI (within-dataset training/evaluation)
MR059	HypeMed [58]	TOIS (2026)	U	MIMIC-III + MIMIC-IV + eICU (DDI omitted on eICU)
MR060	GenRxR [10]	RecSys (2026)	U	MIMIC-III + MIMIC-IV

R4. Appraisal subset and inherited coding

The earlier S1 Table S5 contains 51 rows: 50 numeric ratings and one diagnosis-only N/A row. Excluding its three pre-2022 numeric rows leaves 47 eligible original-roster studies with ratings. The prior scores are retained as supplied; no new reviewer independence, adjudication, prespecification history, or validated risk-of-bias procedure is inferred. Table R2 and `appraisal_subset_47.csv` document the subset underlying the revised manuscript’s appraisal percentages.

C1: MR task definition; C2: dataset/preprocessing; C3: architecture/training; C4: predictive evaluation; C5: safety-aware evaluation; C6: interpretability/evidence/explanations; C7: external validation/robustness/deployment. Ratings are 2 (adequate), 1 (partial), 0 (inadequate/not reported). The broad C5–C7 criteria do not identify whether a particular DDI, contraindication, calibration, hallucination, or external-validation endpoint was measured. In particular, C7 is not a count of external validation.

Table R2: Retained study-level ordinal appraisal (47 studies).

ID	Study/model	C1	C2	C3	C4	C5	C6	C7	Total
MR001	PharmaLLM	1	2	2	1	1	0	1	8
MR003	Confidence calibration	2	1	2	2	1	0	2	10
MR005	FLAME	2	2	2	2	2	1	2	13
MR006	GPSRec	2	2	2	2	2	1	2	13
MR007	KRAM	2	1	1	1	0	0	2	7
MR009	VITA	2	2	2	2	2	2	2	14
MR010	StratMed	2	2	2	2	2	2	2	14
MR011	CausalMed	2	2	2	2	2	2	1	13
MR012	Trans-GAHNet	2	2	2	2	2	1	1	12
MR013	DGCL	2	2	2	2	2	2	2	14
MR014	MR-DTR	2	2	2	2	2	1	2	13
MR015	CIDGMed	2	2	2	2	2	2	1	13
MR016	TreatRAG	2	2	2	2	1	2	2	13
MR017	EDRMM	2	2	2	2	2	1	1	12
MR018	DNMDR	2	2	2	2	2	2	1	13
MR020	TEMPT	2	2	2	2	0	0	2	10
MR021	LEADER	2	2	2	2	0	0	1	9
MR022	SHAPE	2	2	2	2	2	1	1	12
MR023	DKINet	2	2	2	2	2	2	2	14
MR024	MedAlign	2	2	2	2	2	1	2	13
MR025	PAUP	2	1	2	2	2	1	2	12
MR026	MedGCN	2	2	2	2	0	1	2	11
MR027	ACDNet	2	2	2	2	2	1	1	12
MR029	KindMed	2	2	2	2	2	1	2	13
MR030	EGNet	2	2	2	2	2	2	2	14
MR031	KATMed	2	2	2	2	2	1	2	13
MR032	RAG-CPMF	2	2	1	2	2	2	1	12
MR033	TAHDNet	2	2	2	2	1	2	1	12
MR034	CEHMR	2	2	2	2	2	2	1	13
MR035	4SDrug	2	2	2	2	2	1	2	13
MR036	CAMeR	2	1	1	2	2	2	2	12
MR037	EXCERF	2	2	2	2	2	2	2	14
MR038	ARMR	2	2	2	2	0	1	1	10
MR039	TAKECare	2	2	2	2	2	1	2	13
MR040	PROMISE	2	2	2	2	2	1	2	13
MR041	COGNet	2	2	2	2	2	1	0	11
MR043	MoleRec	2	1	2	2	2	2	1	12
MR044	AKA-SafeMed	2	2	2	2	2	1	1	12
MR045	SSPNet	2	2	2	2	2	1	1	12
MR047	KEDRec-LM	2	2	1	1	0	1	1	8
MR048	MedRec	2	2	2	2	0	2	2	12
MR049	KEHGCN	2	2	2	2	2	2	2	14
MR050	BH3-MedRec	2	2	2	2	2	2	2	14
MR052	RAREMed	2	2	2	2	2	0	2	12
MR053	MKAF-Net	2	2	2	2	1	1	2	12
MR054	TCM-KLLaMA	2	1	2	2	1	1	1	10
MR055	DAI-Net	2	1	2	2	2	2	2	13

Available ratings are missing for the following 13 included studies: REFINE, context-aware MR (Chen et al.), HI-DR, IMDR, BiMoRec, SDRBT, HKRec, LAMO, RPNet, Nephrotoxic medication RL, MetaDrug, HypeMed, GenRxR. The five revision additions have descriptive evidence extraction below but no newly assigned ordinal appraisal.

R5. Endpoint-level extraction for the five additions

These findings are based on inspected full texts and must not be extrapolated to the full 60-study register. “Not reported” refers to the inspected source. Four of the five additions report DDI rate; the RL study instead evaluates adverse-outcome discrimination. None of the five establishes prospective clinical harm reduction. Only GenRxR reports a licensed expert’s blinded assessment, concerning synthetic data.

- **RPNet:** clinical-code matching pretraining and rare-code-weighted similar-visit retrieval; binary cross-entropy, multilabel, and DDI objectives. MIMIC-III/IV F_1 : 0.7004/0.6231; DDI rate: 0.0737/0.0723. Test-code deletion evaluates noise sensitivity. Case interpretation does not constitute expert clinical validation; probability calibration, explicit contraindication outcomes, and prospective safety are not reported. Source: publisher PDF pp.6–10, Tables 2–3.
- **Nephrotoxic medication RL:** nine medication actions, offline RL with dead-end discovery, and integrated-gradient feature attributions. Endpoints include mortality, stage-3 AKI, renal replacement therapy, and septic shock. ICU-mortality IQL AUROC: 0.852 internally, 0.845 in a later same-hospital cohort. This is discrimination and temporal validation, not deployed-policy outcome benefit or external institutional validation. No formal probability-calibration or DDI-rate endpoint is identified. Source: Sections 2–4 and Appendix A.
- **MetaDrug:** patient/visit Transformers, self/peer patient adaptation, and uncertainty filtering of support visits. MIMIC-III and Kansas AKI are trained/evaluated within each dataset. DDI rate and code-diversity subgroups are evaluated, but the support-set filter is not output probability calibration. Clinical safety, hard contraindication evaluation, and locked-model institutional transfer are not established. Source: arXiv:2601.22820v1, Sections II–III and Tables III–VI.
- **HypeMed:** knowledge-aware hypergraph pretraining and similar-visit retrieval with longitudinal fusion. Three retrospective benchmarks are evaluated separately. DDI loss/rate apply to MIMIC-III/IV; eICU omits them because the DDI graph is unavailable. The multicenter eICU benchmark therefore adds predictive evidence without a corresponding safety endpoint. No expert recommendation-safety assessment or probability-calibration result is reported. Source: inspected 28-page PDF, Sections 5.1–5.9.
- **GenRxR:** GPT-4o creates counterfactual records; LLaMA 3.1 8B Instruct is adapted for recommendation. DDI and rare/frequent medication-group metrics are evaluated. One licensed expert distinguishes 50 original from 50 synthetic records in a blinded plausibility test. This does not assess clinical safety of recommended regimens, systematic rationale faithfulness, or causal treatment effects. Drug–disease conflict screening is listed as future work. Source: inspected 9-page PDF, Sections 5.1–5.5.

`revision_evidence_extraction.csv` separates DDI, contraindication, calibration, robustness, grounding/hallucination, validation, and expert evidence. Full-corpus endpoint frequencies cannot be recovered from the ordinal appraisal matrix and require a new extraction pass.

R6. Source uncertainties and work required before resubmission

The submitted and later manuscripts report different screening-stage allocations despite the same original 56-study total. The earlier submitted version reports 32 title/abstract exclusions, 61 full texts and five full-text exclusions; the supplied updated source reports 37 title/abstract exclusions, 56 full texts and no original full-text exclusions. Original record-level logs are required to reconcile this history; the revision diagram preserves the updated source’s history and explicitly separates the new scope correction.

Author records are also needed to confirm which eligibility, extraction, taxonomy, and appraisal criteria were prespecified, the independent appraisal-reviewer process, disagreement resolution, and complete database execution details/PRISMA checklist. The original screening-reviewer description is not new evidence of independent appraisal.

The uploaded RL paper has inconsistent cohort totals between Section 2.1 and Appendix Table A.5 and some inconsistent confidence-interval entries. Cohort totals and problematic intervals are therefore not reproduced. GenRxR’s DDI values require table/prose consistency checking before numerical quotation. FLAME names different Aloe checkpoints in the main text and Appendix C; the illustrative LLM table flags this. Full appraisal of the 13 unrated studies, full-corpus endpoint counts, and a complete LLM/RAG evidence table remain outstanding. The original S1 search documentation should be synchronized with this addendum once the authors confirm the historical logs.

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